A Blockchain-Based Uniform e-KYC System for Banks in Bangladesh
Senior Design Project

By

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DECLARATION

It is to declare that this project is our original work. No part of this work has been submitted elsewhere, partially or entirely, for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor.

Relevant previous works presented in this report have been adequately acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

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Abstract

Banking is a formidable force in driving the economy throughout the world, even in Bangladesh. Customer onboarding is a crucial step in a bank's service lifecycle, conducted through Know Your Customer (KYC) policies. Although Bangladesh Bank provides a guideline for KYC, banks often add their own requirements. As a result, whenever a customer wants to open an account at a different bank, they must go through the full KYC process again. This repeated verification is time-consuming for both customers and banks, discouraging people from opening accounts at multiple institutions. Our proposed solution addresses these problems through an e-KYC platform built on smart contracts. Once verified at one bank, a customer does not need to be re-verified elsewhere, as the smart contract securely holds their records and onboarding history. With proper consent, banks can access the relevant information, ensuring privacy. Smart contracts are also tamper-proof, providing proof of verification and safeguarding against fraud. To validate the system's scientific and operational viability, we provide a formal verification using BAN Logic and a comprehensive STRIDE threat model, ensuring the protocol is resilient against spoofing and unauthorized data disclosure. Performance evaluation on the Sepolia testnet demonstrates high efficiency. Overall, our system offers a cost-effective, transparent, and scalable alternative to legacy manual systems, significantly enhancing financial inclusion and operational integrity in the Bangladeshi banking sector.

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Chapter 1

Introduction

1.1 Background and Motivation

Know Your Customer (KYC) procedures in the present financial landscape have since developed into national security tools rather than compliance measures that were formerly carried out as a matter of routine [1]. Although movement of funds across countries is almost real time and completely digital, the system of identity verification is still mostly archaic, manual, and disintegrated [2, 3]. This disproportion puts a significant international burden on the Anti-Money Laundering (AML) and Countering the Financing of Terrorism (CFT) models [4]. Despite the fact that several jurisdictions have started to move out of these legacy systems, Bangladesh is significantly stuck and still uses traditional KYC mechanisms which do not satisfy the contemporary requirements [5].

The operational costs increase is not the only effect of the paper-based KYC procedures on the congestion [6]. These mechanisms act as the major barriers to financial inclusion where millions of people cannot access formal banking services [7]. The implicit cost imposed on the customer in the form of unnecessary time demands, complexity of documents, and even direct financial costs is an aversion to involvement in the regulated financial ecosystem [8]. Although a single regulatory framework is in place, with the Bangladesh Financial Intelligence Unit (BFIU) instituting the rule, which is implemented under the Money Laundering Prevention Act, 2012, the real application of an interoperable and comprehensive KYC infrastructure is still divided [9]. A customer who is verified by a single financial institution ("Bank A") is considered to be a totally unverified customer by another ("Bank B"). This lack of interoperability leads to widespread redundancy, forcing both financial institutions and customers to repeatedly undergo the same rigorous verification procedures. These manual processes also inject latency, and further amplify the chances of data duplication and the risk of identity fraud [10]. They also worsen the general experience of both the customers and the bank officials, as far as usability is concerned.

1.2 Purpose and Goal of the Project

The current efforts of dealing with these inefficiencies can be broadly divided into two groups, both of which have significant drawbacks. Centralized KYC databases are a single point of failure, and therefore are an easy target of cyberattacks, and prone to systemic corruption [11, 12]. Conversely, blockchain-based identity solutions, such as the Self-Sovereign Identity (SSI) models [13, 14], focus on user autonomy but often lack the central oversight required by local mandates. Although these methods are technologically advanced, they often ignore local legal demands of central bank control, which makes them inconsistent with the Anti-Terrorism Act, 2009 [4, 15]. As a result, there is an evident lack of structures that can adequately integrate the cryptographic assurances of blockchain with the regulatory power that is needed in emerging economies. Operationally, the present KYC procedures are largely based on manual documentation or direct database manipulation and provide poor guarantees of safety, integrity, or trust [6, 10]. It is worth noting that the notion of self-sovereign identity is practically nonexistent in the current banking processes, yet millions of transactions and account openings are carried out daily under these insecure paradigms.

In order to solve these problems, this paper suggests a hybrid electronic KYC (e-KYC) system based on Ethereum smart contracts. The suggested architecture is based on a semi-decentralized consortium model in which the central bank acts as the *root of trust* and holds overall administrative control over the system, while identity data storage is distributed across multiple nodes to increase system resilience and avoid a single point of failure. The fundamental aspect of the framework is the "verify once and reuse everywhere" mechanism, which allows financial institutions to use verified identity credentials through cryptographic hashing without disclosing sensitive personal information. The proposed system enables banks to share a trusted identity layer and maintain complete control over their internal risk models, transaction limits, and operating policies by decoupling identity verification from financial transaction processing. Such a design guarantees regulatory compliance without compromising institutional independence.

1.3 Organization of the Report

This paper presents an end-to-end proposal for a blockchain-based e-KYC system specific to the Bangladeshi regulatory environment. The main contributions of this work are as follows:

1. We have created a privacy-preserving architecture that decouples on-chain identity hashes from off-chain data storage, ensuring data integrity while satisfying privacy requirements.

2. Formal security verification using STRIDE threat modeling and BAN logic demonstrates the system's resilience against spoofing, information disclosure, and unau-

thorized access.

3. An empirical evaluation on the Sepolia testnet indicates a 10–30% reduction in gas expenditure relative to current industry benchmarks, supporting the economic viability of local financial institutions.

4. A full-stack functional prototype, including smart contract logic and a front-end interface, illustrates a deployable framework for the financial sector.

Chapter 2

Background and Literature Review

2.1 Preliminaries

KYC nowadays is a crucial requirement for banks worldwide but traditional paper-based KYC processes are slow and prone to human errors. As digital banking is growing, research has been already focusing on electronic KYC (e-KYC) systems that use digital identity and biometric verification to secure data exchange. It also speeds up customer on-boarding while maintaining AML/CFT compliance. However, most of the solutions are still prototypes and they lack the regulatory needed for real deployment especially in developing countries. In Bangladesh, only a few institutions offer e-KYC. But these systems are still isolated. There is no shared or cross-bank mechanism. Hence, it forces customers to repeat verification in every bank. This literature review therefore explores existing e-KYC approaches to identify their strengths and limitations to clarify the gap. Overall, it motivates a unified e-KYC system for Bangladeshi banks.

Over the few years, blockchain has caused a significant shift from centralized databases to user-centric and decentralized models. It has helped to address vulnerabilities such as single points of failure, data breaches and unauthorized access. Centralized systems rely on a single authority for identity management. So they are prone to hacks and manipulation, as proven by numerous incidents in financial sectors. To tackle that, researchers have pivoted toward decentralized architectures, emphasizing Self-Sovereign Identity (SSI) and Decentralized Identifiers (DIDs). SSI empowers users to own and control their digital identities without intermediaries and DIDs provide unique, verifiable identifiers anchored on blockchain ledgers. These methods enable selective disclosure, where users share only necessary attributes. Hence, these greatly enhance privacy of the customers. Techniques like zero-knowledge proofs (ZKPs) allow verification without revealing underlying data. Besides, hashing ensures sensitive information is stored off-chain and maintains on-chain integrity.

2.2 Literature Review

2.2.1 Related Research

For instance, Dong et al. propose a blockchain-based SSI system for KYC processes utilizing DIDs and verifiable credentials. It enables users to selectively disclose identity attributes [13]. This framework enhances privacy through ZKPs and reduces costs by eliminating redundant verifications. The system also ensures AML/CFT compliance via immutable consent recording with real-time regulatory updates. However, it overlooks infrastructure limitations in developing countries like Bangladesh.

Soltani et al. [14] introduce a new approach [14] to client on-boarding using SSI and distributed ledger technology. Here, the users control their credentials on a permissioned blockchain. By leveraging DIDs, the system allows for tamper-proof verification and user-centric data ownership. It preserves privacy through off-chain storage and cryptographic proofs. This model aligns with GDPR-like standards but requires coordinated credential formats and thus poses challenges for fragmented banking ecosystems.

Patkar et al. [16] present a secure and reliable e-KYC system using blockchain. It focuses on user-centric identity with hashing for off-chain data protection. In the system, smart contracts manage consent and access that helps secure data sharing across institutions while reducing exposure. Although the approach emphasizes transparency and efficiency, it does lack the integration with local regulations such as the BFIU guidelines.

Replacing older frameworks, Karmoker et al. describe a decentralized e-KYC system for secure identity verification in Bangladesh. It incorporates SSI principles with hashing and consent mechanisms [15]. This architecture integrates NID-based verification, supports revocation for AML compliance. It also uses DIDs to allow user-owned data, addressing privacy and encouraging cross-bank reuse. However, the framework remains largely conceptual without a detailed deployment blueprint. Besides, it relies on a public Ethereum blockchain that incurs high gas costs and network delays. It also lacks formal security verification methods such as BAN Logic for protocol soundness or STRIDE for comprehensive threat modeling. It does not include performance evaluations on testnets or economic cost projections of the system. Moreover, while it mentions consent, it does not specify cryptographically signed mechanisms to prevent indefinite data access. Also, revocation is not restricted to the original issuer bank which may complicate the legal compliance under the BFIU guidelines.

Fugkeaw [17] proposes a blockchain-based e-KYC framework named “e-KYC TrustBlock” that integrates Ciphertext-Policy Attribute-Based Encryption (CP-ABE) with a permissioned blockchain called Hyperledger Fabric. In this method, sensitive KYC documents are symmetrically encrypted and stored off-chain in the cloud. It enforces access policies and consent rules to be on-chain using CP-ABE. The main strengths of this approach are fine-grained access control, strong privacy preservation, and immutable audit trails.

However, the heavy use of attribute-based encryption introduces some computational overhead. The system depends on reliable cloud storage and a well-established permissioned network that may limit its uses in a resource-constrained environment.

Even with digital tools, banks often operate in isolation. This leads to redundancy and inefficiencies in customer on-boarding. This repeated process wastes both time and resources for banks and customers. Studies estimate that global banks spend over $500 million each year per institution on KYC compliance. Still on-boarding a new customer can take several days on average. This process also increases fraud risks and operational burdens because data duplication across different banks creates vulnerabilities. Therefore, there is a need for one-time identity verification to mitigate these issues. Many researchers are now leveraging blockchain for shared and immutable ledgers, focusing on time and cost efficiency.

Tabassum and Arsheen [11] propose a blockchain-based secure e-KYC solution for banking systems that use distributed ledger technology to introduce one-time verification across multiple banks. This method reduces redundancy by storing hashed data on-chain, cutting on-boarding time by up to 70% and costs through eliminated middlemen. The framework enhances efficiency with smart contracts for automated workflows but overlooks the varying bank policies in countries like Bangladesh.

Patil and Sangeetha develop a blockchain-based decentralized KYC verification framework for banks. Here, institutions vote on data legitimacy via smart contracts [12]. This operational model saves time by enabling quick consensus-based approvals, reducing fraud through collective oversight. It also lowers costs by avoiding repeated checks. However, it introduces coordination overhead for cross-bank networks.

Bhatia et al. [18] outline a cost-efficient blockchain-based e-KYC platform using biometric verification. Their system allows one-time processing with reusable keys. This approach decreases operational costs by eliminating duplicate document submissions and customer on-boarding time. It also preserves privacy through off-chain storage. Its challenges include gaps in biometric infrastructure especially in developing economies.

Karadag et al. [19] present a blockchain-based KYC model for credit allocation in banking. The model allows real-time data sharing after verification. By using a private Proof-of-Stake ledger, it reduces inter-bank on-boarding time and costs through immutable records. It has efficiency in credit decisions as well. However, it has some limitations such as regulatory hurdles and integration with legacy systems.

Gangurde et al. [20] present a practical blockchain-based e-KYC system tailored particularly for the Indian context. It primarily leverages Aadhaar and PAN verification [20]. Once the users are validated, only the verification status and a cryptographic hash are recorded on the blockchain. Thus, other banks can check the customer's KYC status without requesting documents again. This design provides one-time verification, removes

repetition and saves time. Nevertheless, the system is tightly coupled with India's Aadhaar infrastructure and government APIs that makes its adaptation almost impossible in other contexts. It also lacks robust mechanisms for consent management and credential revocation.

2.2.2 Similar Applications

Table 2.1 illustrates a comparative analysis of the similar applications below:

Table 2.1: A Comparative Analysis of Related Works

Author & Year	Blockchain Architecture	Identity Model	Regulatory Oversight?	Revocation Mechanism?	Privacy Preservation	Implementation Scope
Dong et al., 2024	Hybrid	SSI	Partial	Yes	ZKP	Prototype
Soltani et al., 2018	Consortium	SSI	No	Yes	ZKP	Prototype
Patkar et al., 2024	Private	User-Centric	No	No	Hashing Only	Conceptual
Karmoker et al., 2024	Consortium	User-Centric	Yes	Yes	Hashing Only	Conceptual
Tabassum et al, 2025	Private	User-Centric	No	No	Off-Chain Storage	Conceptual
Patil et al., 2022	Public	User-Centric	No	Yes	Hashing Only	Prototype
Bhatia et al., 2024	Private	User-Centric	No	No	Hashing Only	Conceptual
Karadag et al., 2024	Private	User-Centric	Partial	Yes	Hashing Only	Conceptual
Fugkeaw, 2022	Hybrid	User-Centric	No	Yes	ZKP	Prototype
Gangurde et al., 2023	Hybrid	User-Centric	No	No	Hashing Only	Prototype
Our work	Consortium	SSI & User-Centric	Yes	Yes	ZKP	Prototype

2.3 Gap Analysis

Although there are lots of innovative solutions, many of them do not work for Bangladesh as they ignore the authority of the central bank under the BFIU guidelines and the AML/CFT laws [4]. Fully decentralized models focus on giving users complete control over their data. However, they are often criticized because they lack clear mechanisms for regulatory intervention. Without such mechanisms, authorities cannot easily monitor or enforce compliance. Pure SSI systems may not comply with AML requirements because governments cannot easily track or revoke identities of bad actors. It creates a tension between privacy and traceability. Additionally, infrastructure gaps such as limited digital literacy, uneven biometric adoption and unreliable networks hinder deployment in developing countries. Our work bridges the gap by proposing a smart contract-based e-KYC platform that ensures one-time verification, user consent, privacy through hashing and regulatory compliance via BFIU-integrated revocation mechanisms.

Chapter 3

Methodology

3.1 System Implementation

3.1.1 Backend Service Modules

User Registration and Authentication Service

The user registration and the layer of authentication that determines if a user or bank officer is legit is determined with the help of this service. For that role separation is done with the help of RBAC(Role based access control). The authentication service uses JWT(JSON Web Tokens) to validate the legitimacy of every API request.

```
// Security middleware
app.use(helmet({
    contentSecurityPolicy: {
        directives: {
            defaultSrc: ["'self'"],
            styleSrc: ["'self'", "'unsafe-inline'"],
            scriptSrc: ["'self'"],
            imgSrc: ["'self'", "data:", "https:"],
        },
    },
    crossOriginEmbedderPolicy: false,
}));

app.use(cors({
    origin: process.env.CORS_ORIGIN?.split(',') || '*',
    credentials: true,
    methods: ['GET', 'POST', 'PUT', 'DELETE', 'PATCH', 'OPTIONS'],
    allowedHeaders: ['Content-Type', 'Authorization']
}));

```

Figure 3.1: This code snippet shows the API hardening baseline using helmet (secure HTTP headers, CSP directives) and CORS (controlled cross-origin access), which protects backend endpoints from common web attack vectors while allowing frontend integration.

The main reason for JWT is that the token based approach will contain the relevant user information in the token itself rather than using a database to store the information. This aligns with our goal of not using off chain databases on a large scale. The way we implemented the token based approach is in two folds. One is the short lived token that has an expiry of 15 minutes. And the other is the long refresh token which has an expiry range of a week . The long refresh token is stored in redis for secure session reloading and blacklisting whenever it is needed. The implemented middleware layers of the system are integrated with express rate-limit to mitigate denial of service attacks.

```
// Rate limiting - completely disabled in test environment
if (process.env.NODE_ENV !== 'test') {
    const limiter = rateLimit({
        windowMs: (parseInt(process.env.RATE_LIMIT_WINDOW) || 15) * 60 * 1000,
        max: parseInt(process.env.RATE_LIMIT_MAX_REQUESTS) || 100,
        message: 'Too many requests from this IP, please try again later.',
        standardHeaders: true,
        legacyHeaders: false,
    });
    app.use('/api/', limiter);
}

```

Figure 3.2: This code enforces request throttling at API level to mitigate brute-force and denial-of-service behavior by limiting requests per IP within a configurable time window

The HS256 algorithm ensures integrity of the tokens. The overall session management is done in O(1) time complexity.

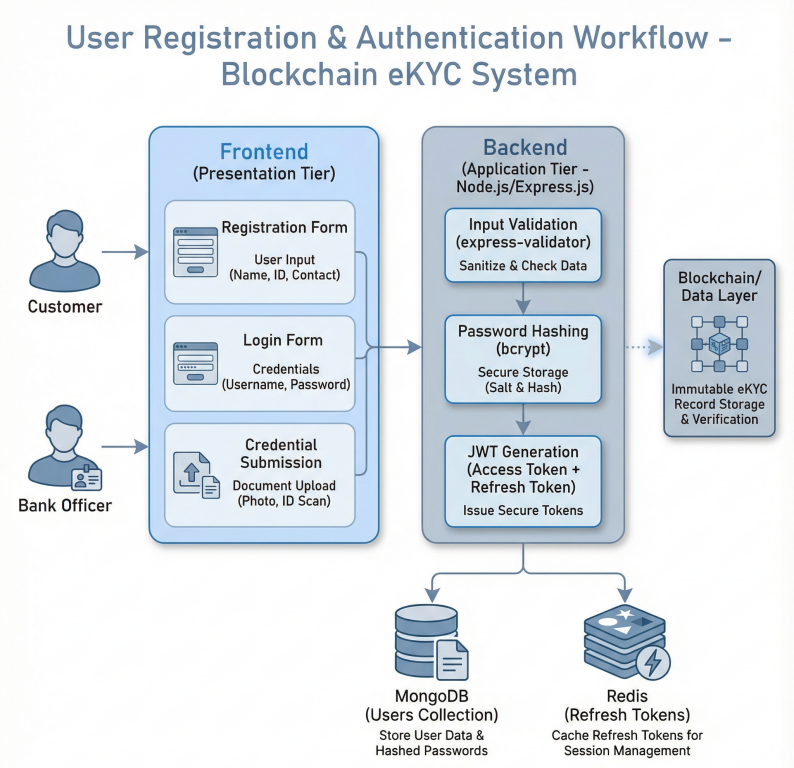


Figure 3.3: Illustration of the user registration and authentication workflow also showing the detailing input validation using the express-validator library, password hashing using bcrypt and the generation of JWT access along with refresh tokens for secure session management.

KYC Submission Service

The KYC submission service is the main endpoint where all the required documents are needed for the first hand verification. As already mentioned before the goal of the system is to make sure that verification is done once and then for the recurring use cases this verification proof will act as the enabler for doing further account openings in the same bank or in other banks within the ecosystem. This enablement is done through the aforementioned service. Along with that, this service also makes sure that the user adhered to the requirement document submission flow before the relevant document copies reach the bank officer end.

```
const createKYCRequest = async (data) => {
    try {
        const {
            userId,
            bankId,
            nidNumber,
            nidData,
            requestType = 'new',
            kycLevel = 1,
            useExistingCredential = false,
            sharedCredentialId = null
        } = data;

        // Validate required fields
        if (!userId) {
            throw new Error('User ID is required');
        }

        if (!bankId) {
            throw new Error('Bank ID is required');
        }

        if (!nidNumber && !useExistingCredential) {
            throw new Error('NID number is required');
        }

        // Validate NID format only if not using existing credential
        let nidValidation = { valid: true, format: 'shared-credential' };
        if (!useExistingCredential) {
            nidValidation = verifyNIDFormat(nidNumber);
            if (!nidValidation.valid) {
                throw new Error(nidValidation.message);
            }
        }

```

Figure 3.4: This snippet demonstrates backend-side validation gates for KYC submission (required identifiers, NID constraints, and flow branching), ensuring malformed or incomplete requests cannot enter the workflow.

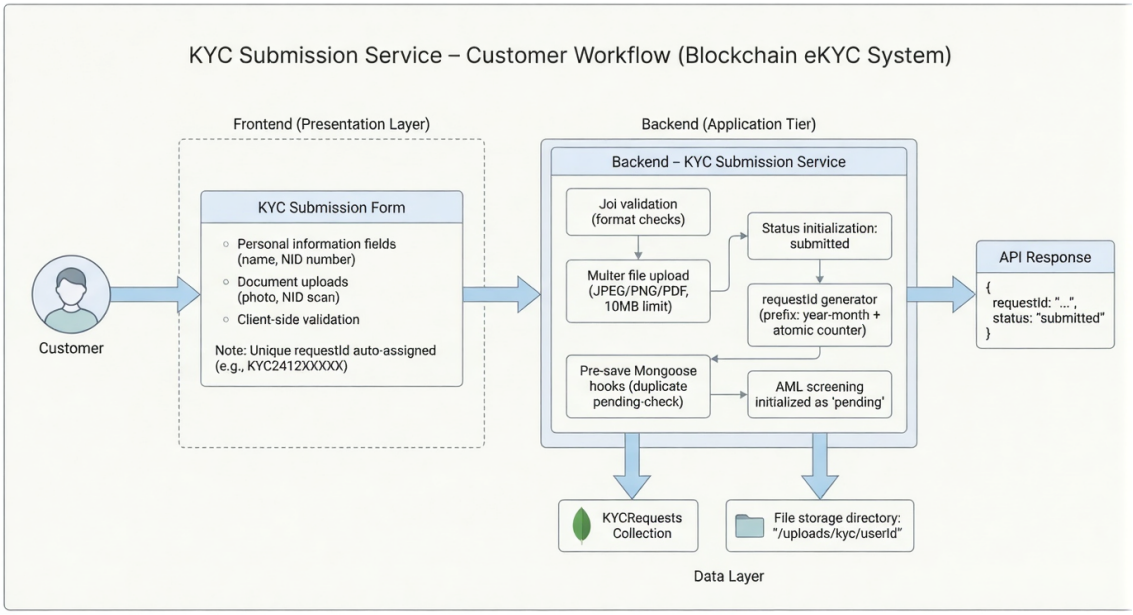


Figure 3.5: Illustration of the customer's KYC submission workflow highlighting document upload stages (photo, NID scan), Joi validation for format checking, AML screening initialization and the generation of review request for the bank officers

The submitted document data are validated using schemas that are extracted from authorized legitimate sources. After validation is done the data is stored in a secured offchain database (mongoDB in our case) so that later on these data can be used for on chain verification without exposing the confidential data. The reason for not keeping the sensitive data on the blockchain is due to information leakage. If on-chain platform is used for storing sensitive data then all nodes can see the data and the point of privacy gets void at that point. This service generates a cryptographic hash using SHA256 and keeps it on the on chain platform as a zero knowledge proof that the user has verified their documents successfully.

KYC Verification and Approval Service

This service actively verifies submitted KYC documents of users from the perspective of a bank officer. Upon this bank officers can review KYC requests and based on the set requirements the bank officers can accept or reject these reviewed KYC requests. Considering these, this service is pivotal in shaping the structure before the issuance of the on-chain credential.

```
case 'rejected':
    kycRequest.workflow.rejectedAt = new Date();
    kycRequest.workflow.reviewCompletedAt = new Date();

    // Set rejection details
    if (options.rejectionCategory) {
        kycRequest.rejectionReason = {
            category: options.rejectionCategory,
            details: options.rejectionDetails || '',
            canReapply: options.canReapply !== false,
            reapplyAfter: options.reapplyAfter || new Date(Date.now() + 30 * 24 * 60 * 60 * 1000)
        };
    }

    // Update user KYC status
    const rejectedUser = await User.findById(kycRequest.userId);
    if (rejectedUser) {
        rejectedUser.kycStatus = 'rejected';
        await rejectedUser.save();
    }
    break;

case 'issued':
    // This status is set after blockchain credential issuance
    kycRequest.blockchain.issuedAt = new Date();

    // Update user
    const issuedUser = await User.findById(kycRequest.userId);
    if (issuedUser) {
        issuedUser.kycStatus = 'approved';
        await issuedUser.save();
    }
```

Figure 3.6: This section implements decision-state transitions (approved, rejected, issued) and synchronizes workflow timestamps and user KYC status, forming the core policy logic of officer review outcomes.

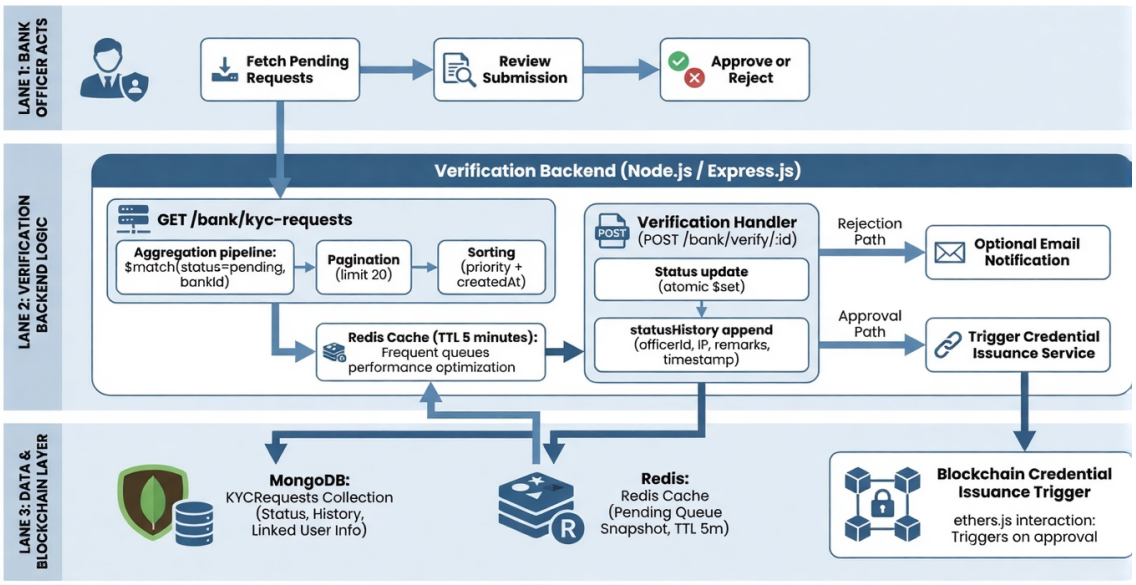


Figure 3.7: Logic for KYC verification and approval from bank officer action to backend system and the final trigger for blockchain credential issuance upon validation

Moreover, there are intuitive headers provided to help the bank officers accept or reject the KYC requests. The request state is directly updated based on the verification decision. If the KYC request was approved then the Blockchain Service is triggered to issue a credential on the ledger. And if the request was rejected then the lifecycle will be terminated from the system.

Dashboard and Monitoring Services

This service offers a dashboard to track role specific visibility which supports transparency and visual cue for the customers and bank officers. This service is strictly for monitoring purposes and does not play around with the underlying system state.

```
// Health check endpoint
app.get('/health', (req, res) => {
    res.status(200).json({
        status: 'healthy',
        timestamp: new Date().toISOString(),
        uptime: process.uptime()
    });
});
```

Figure 3.8: This health-check endpoint exposes runtime service status and uptime for monitoring, deployment verification, and incident triage support.

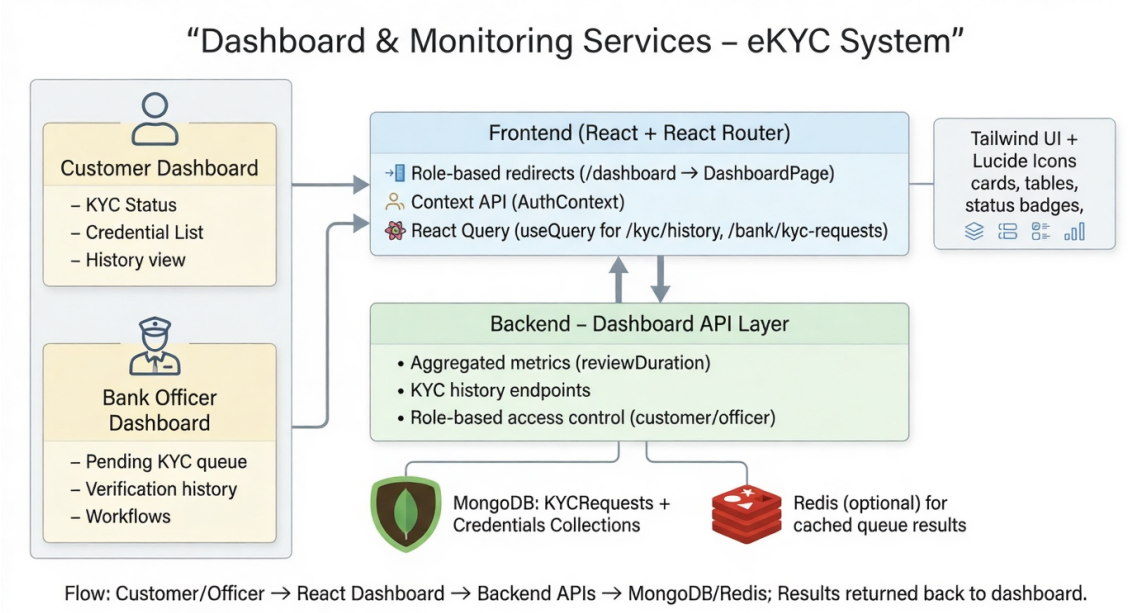
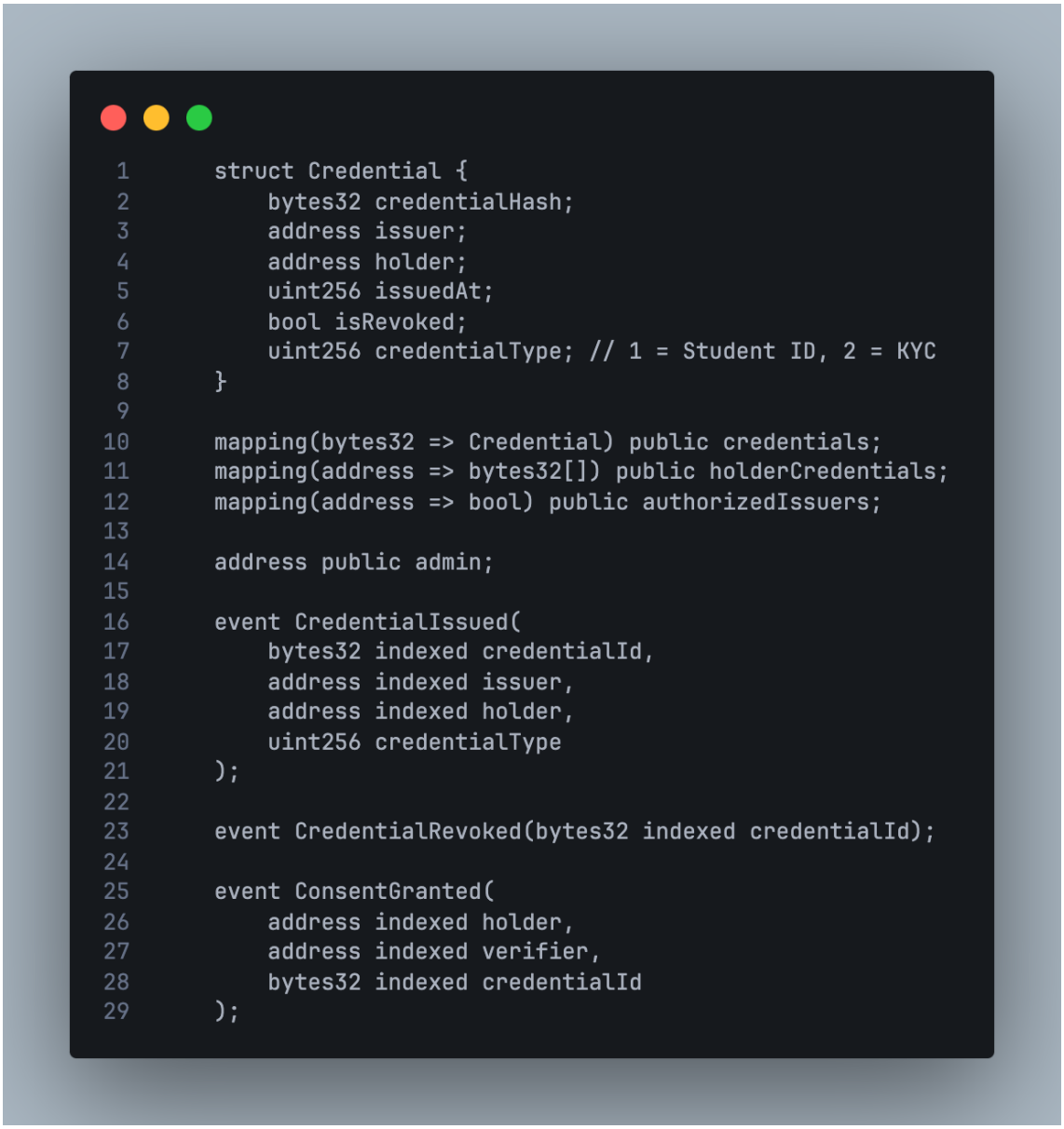


Figure 3.9: System monitoring dashboard interface for customers and bank officers showing the role-based views for KYC status, credential lists and aggregated metrics such as: review duration, pending KYC queue, kYC status etc depending on the role

3.1.2 Blockchain Integration Layer

Blockchain Based Credential Issuance and Management

The integrated blockchain layer in our system serves as a shared tamperproof registry for verifiable identity credentials. These identity credentials follow a set lifecycle where it could be issued, verified and revoked.



```
struct Credential {
    bytes32 credentialHash;
    address issuer;
    address holder;
    uint256 issuedAt;
    bool isRevoked;
    uint256 credentialType; // 1 = Student ID, 2 = KYC
}

mapping(bytes32 => Credential) public credentials;
mapping(address => bytes32[]) public holderCredentials;
mapping(address => bool) public authorizedIssuers;

address public admin;

event CredentialIssued(
    bytes32 indexed credentialId,
    address indexed issuer,
    address indexed holder,
    uint256 credentialType
);

event CredentialRevoked(bytes32 indexed credentialId);

event ConsentGranted(
    address indexed holder,
    address indexed verifier,
    bytes32 indexed credentialId
);
```

Figure 3.10: This snippet defines the on-chain credential data model and event schema. It is the foundational registry structure used for lifecycle tracking and auditability.

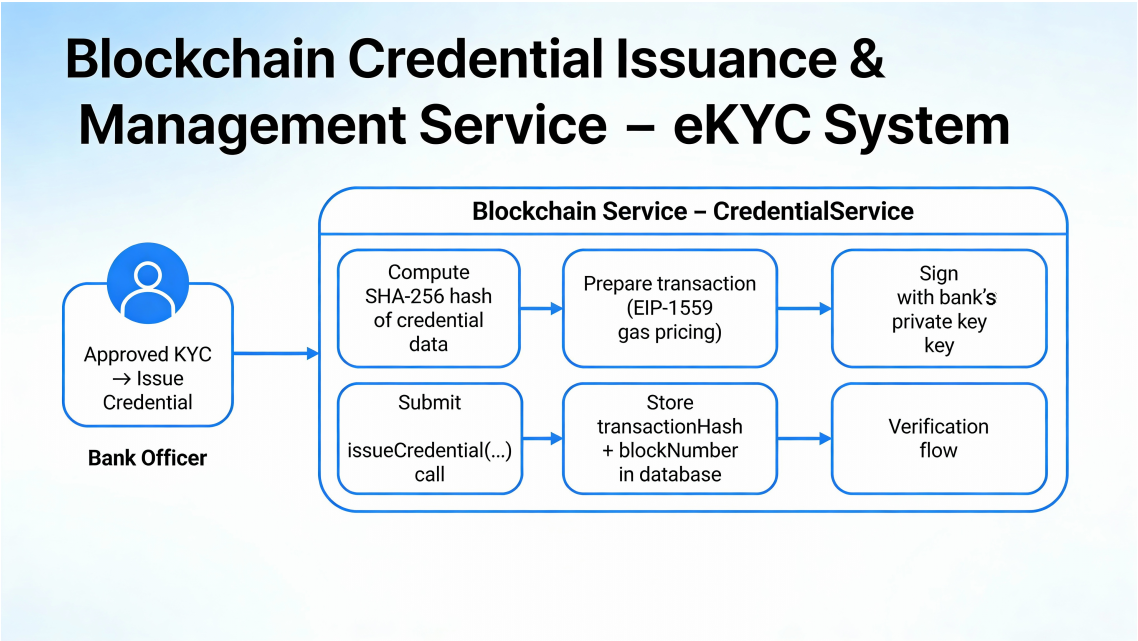


Figure 3.11: Credential Lifecycle Management outlining the steps for hashing credential data with SHA-256, signing transactions with bank private keys and updating the on-chain registry for keeping records for future use cases

The state management is done by a registry style smart contract that enforces strict access control. The act of issuance or revocation can only be done by an authorized bank. Moreso, verification remains as a read only operation which is accessible to an authorized verifier(bank officer) . We implemented revocation in such a way that once a credential is revoked that will resonate throughout the ecosystem. This ensures compliance if any suspicious activity is found.

```
function verifyCredential(bytes32 _credentialId, bytes32 _hash)
    public
    view
    returns (bool)
{
    Credential memory cred = credentials[_credentialId];
    return cred.credentialHash == _hash && !cred.isRevoked;
}

function revokeCredential(bytes32 _credentialId) public {
    Credential storage cred = credentials[_credentialId];
    require(msg.sender == cred.issuer, "Only issuer can revoke");
    cred.isRevoked = true;
    emit CredentialRevoked(_credentialId);
}

function grantConsent(address _verifier, bytes32 _credentialId) public {
    require(credentials[_credentialId].holder == msg.sender, "Not credential holder");
    emit ConsentGranted(msg.sender, _verifier, _credentialId);
}
```

Figure 3.12: These functions implement verification checks, issuer-controlled revocation, and holder-driven consent signaling, covering the complete trust-control surface of credential usage.

On-Chain Consent Management Mechanism

The main idea for implementing this mechanism was so that no third parties like bank officers can access credentials without the user knowing. With this mechanism we are maintaining explicit user authorization as a first class condition. If there is no consent then the bank officers can not proceed with any sort of verification or data scrutiny. This layer makes sure that data is protected for every user and upon their will only the bank officer can get to work with the data.

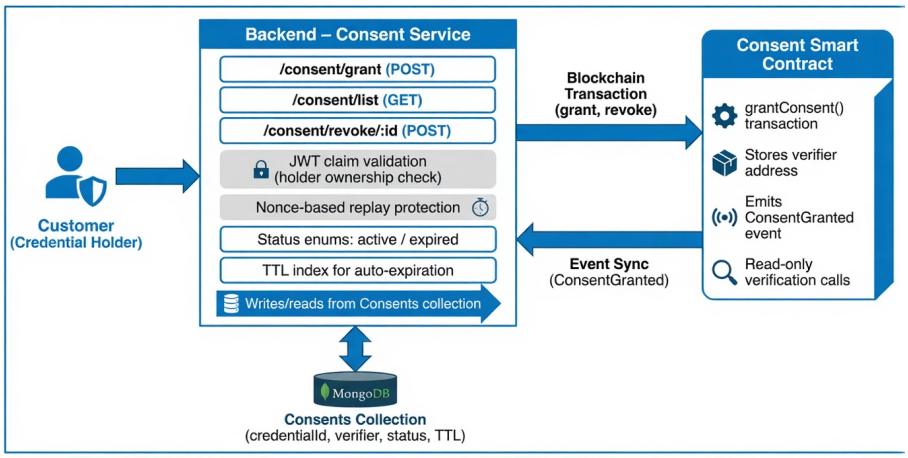


Figure 3.13: User-centric consent management flow showing how customers grant or revoke access to their credentials through smart contract transactions

This mechanism records consent in an immutable manner which binds a holder to a verifier for a limited time. We designed this consent workflow in a way which is time bounded and also revocable. This setting act as a gatekeeper for the off-chain systems before they can get hold of any data.

3.1.3 System Interaction Flow

Customer Interaction Flow

The customer's journey starts with their registration . After registration they are authenticated through the establishment of a secure session by JWT. As shown in **Figure 9**, the user access the portal to start the KYC process. The data submission phase **Figure 10** and **Figure 11** allows users to submit their Personally Identifiable Information like NID numbers and addresses . These datapoints are then verified using the acquired schemas from authorized sources.

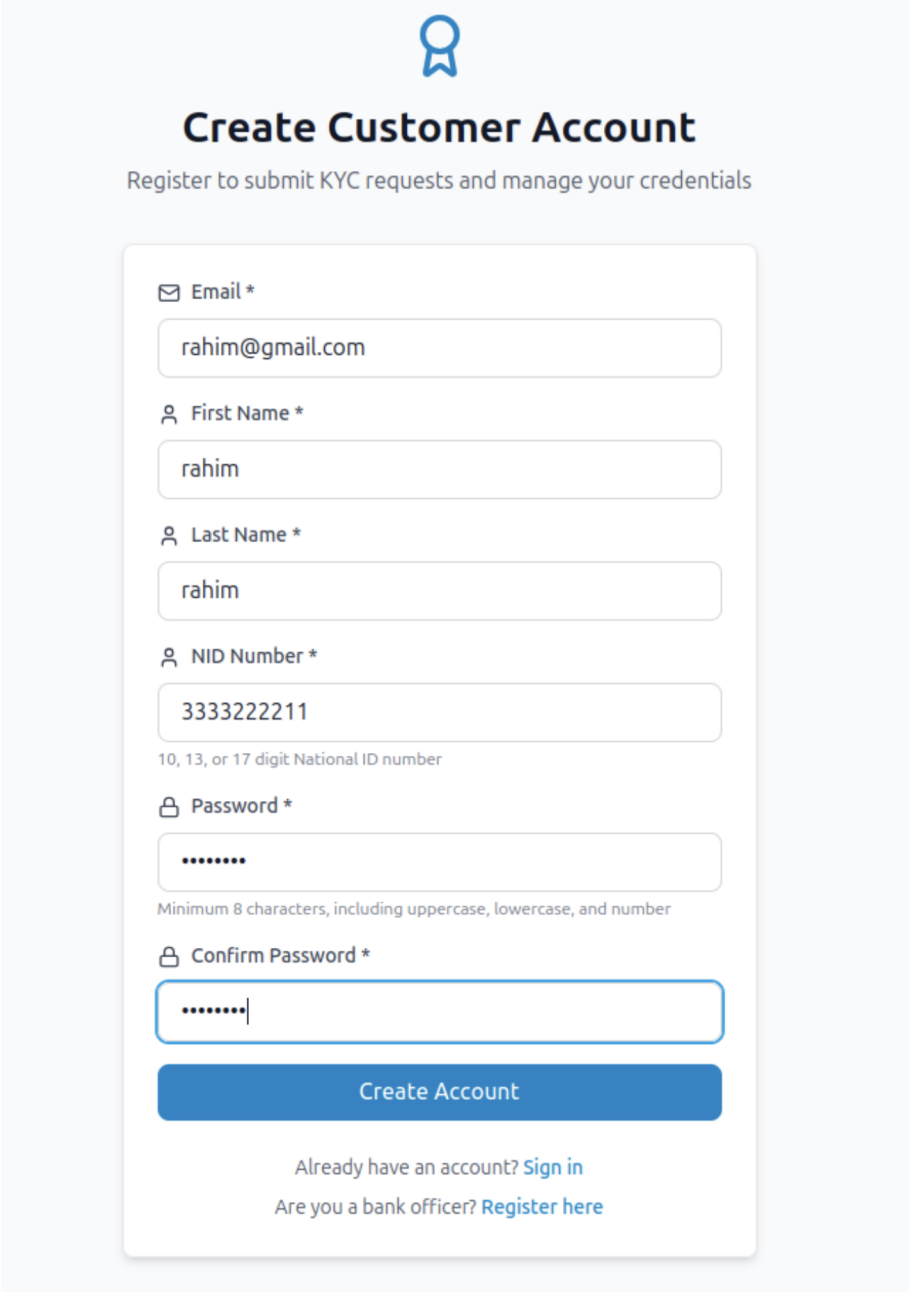


Figure 3.14: Customer account creation and authentication portal interface using which users can register themselves in the system with email, NID details and password

The system generates a cryptographic hash of the uploaded documents as shown in the document uploading and hashing phase **Figure 10** and **Figure 11**. This establishes the fact that only the hash is required for the on chain verification not the uploaded documents. Once the bank officer reviews and accepts the process the user can then see their credential as shown in **Figure 11**.

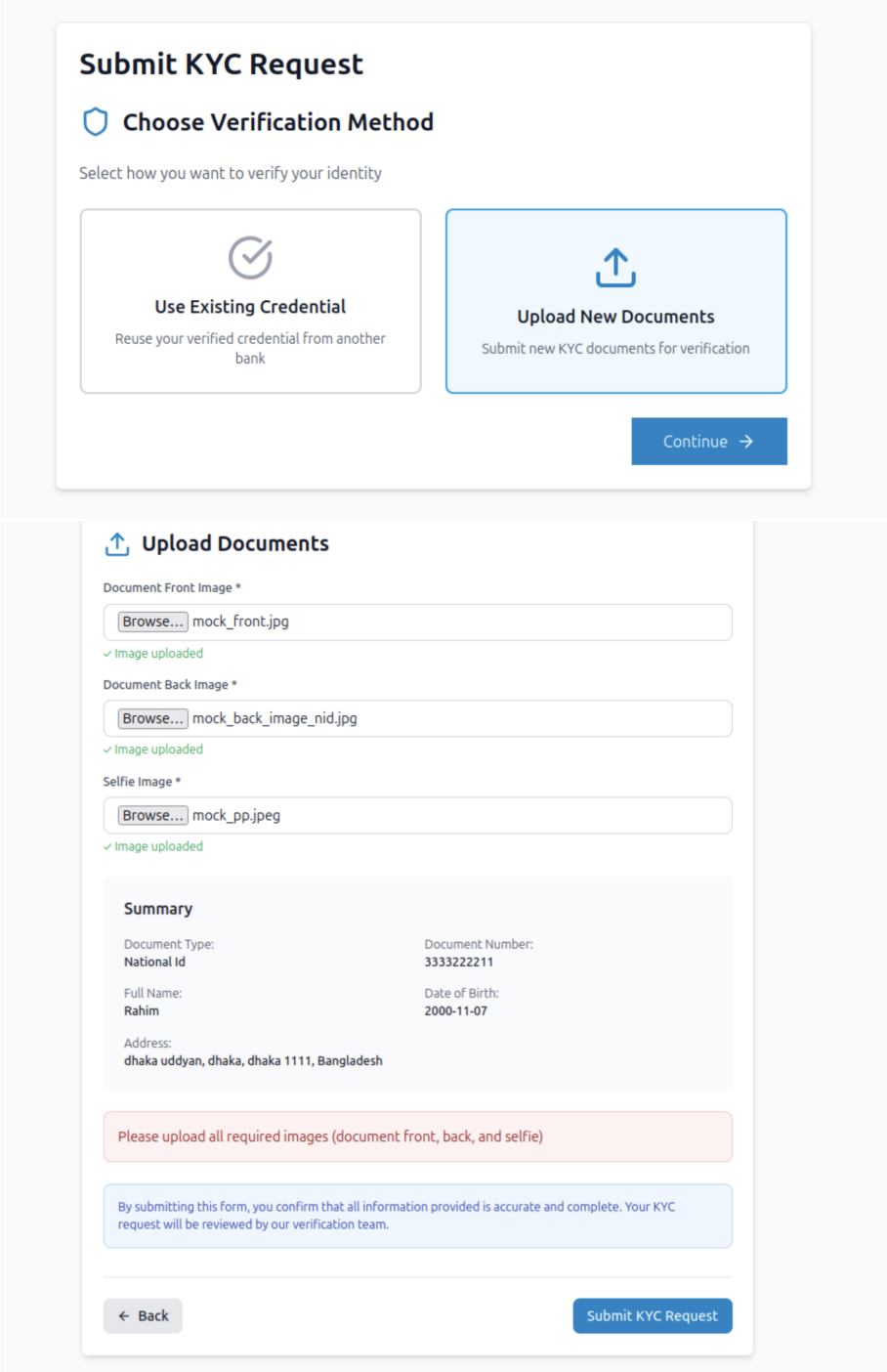


Figure 3.15: Interface for customer document submission using which users can choose between using existing credentials or upload new NID images and selfies for cryptographic hashing

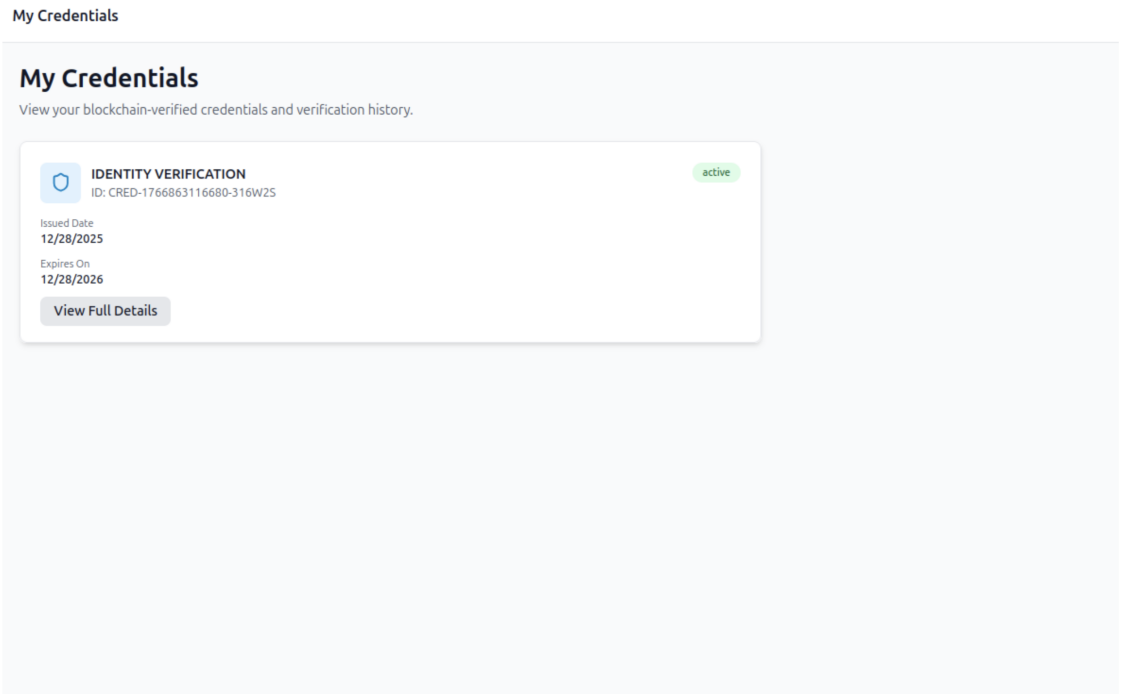


Figure 3.16: Credentials wallet view for customers through which customers can see their active blockchain-verified identity credentials with unique IDs generated by the blockchain service with the help of the backend along with issuance dates and expiry information of that credential

Bank Officer Interaction Flow

The officer dashboard as shown in **Figure 13** provides a real time work queue of pending applications. Officers perform a detailed risk and document review which assess the customer's profile against regulatory standards.

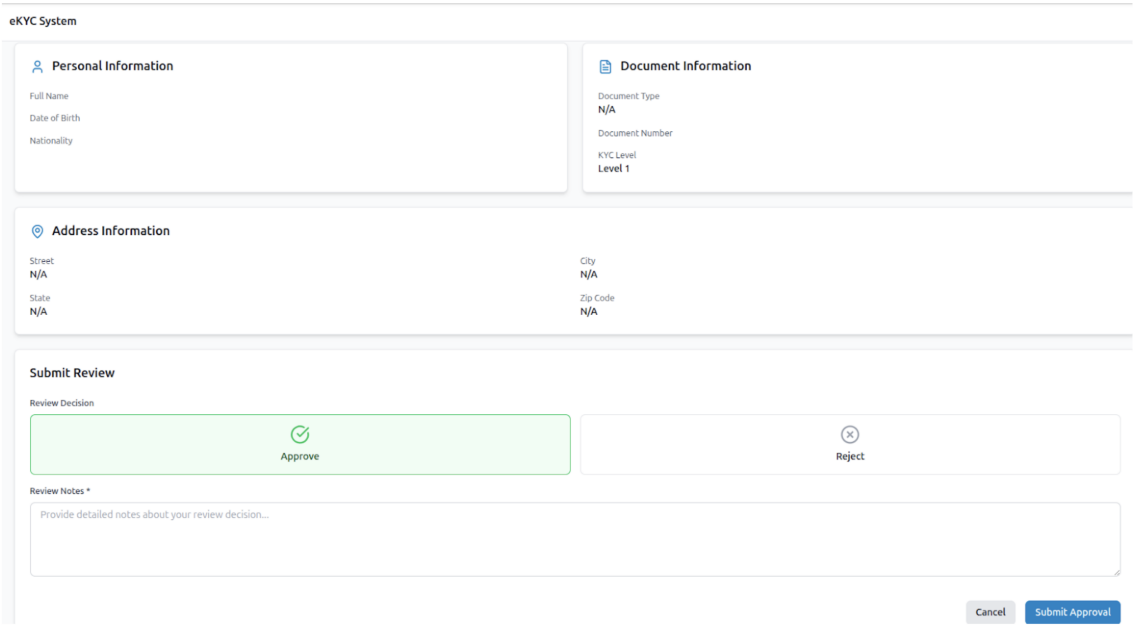


Figure 3.17: With this decision management module, bank officers can approve or reject KYC submissions based on detailed personal, address, historic data and document information review

As shown in **Figure 12**, the decision management module allows the bank officer to either approve or reject a request. An approved request automatically triggers the blockchain credential issuance phase. There the system interacts with the credential registry contract to mint the digital certificate. This record is permanent but can be revoked **Figure 14** by a bank officer if there arise events which violate AML/CFT policies by the credential holder.

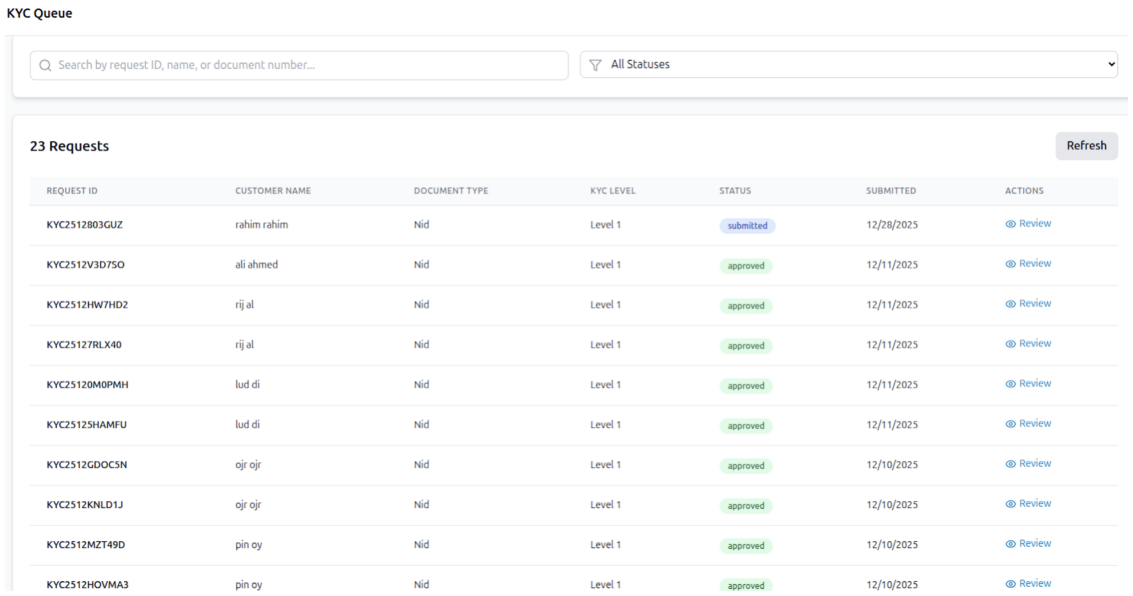


Figure 3.18: Bank officer view for seeing pending, approved or rejected KYC requests along with document type of the submission, submission date and status etc

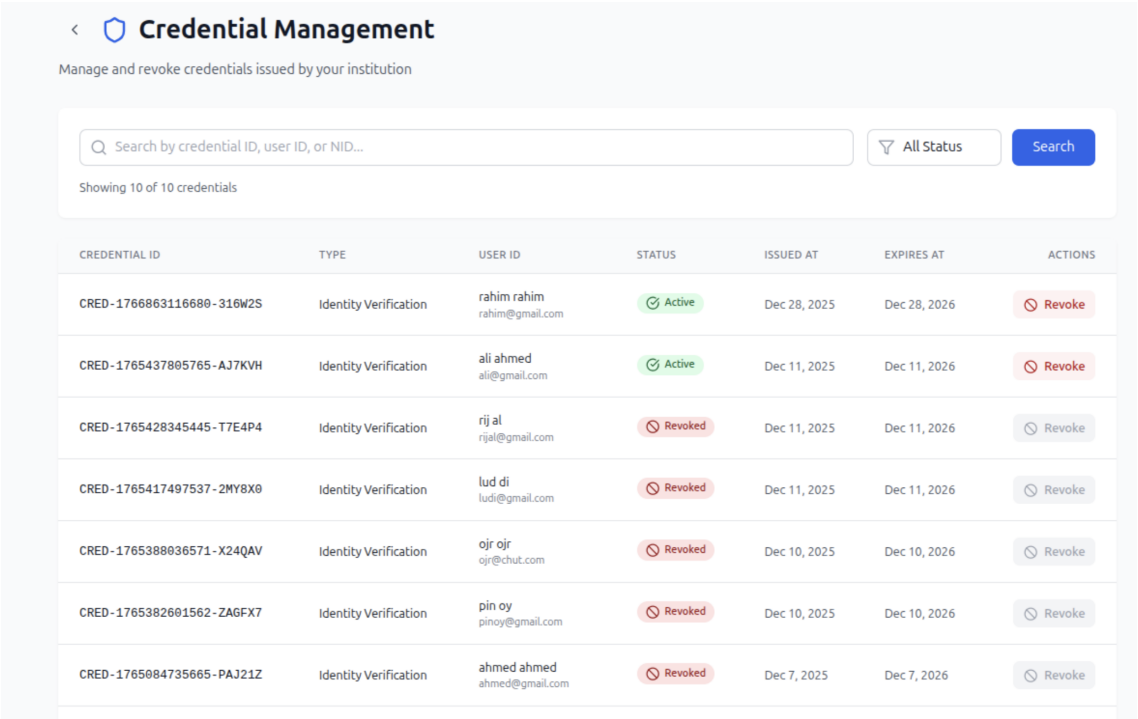


Figure 3.19: The Credential Management dashboard interface displays the status, credential issuance details, and revocation controls

Chapter 4

Investigation/Experiment, Result, Analysis and Discussion

4.1 Design Philosophy and System Workflow

The proposed architecture adopts a semi-decentralized consortium model where the Central Bank serves as the root of trust within a specific environment. The system is built upon a "Verify Once, Reuse Everywhere" identity model which fundamentally decouples the off-chain process of identity verification from the on-chain mechanism of access authorization. Meaning, the process forwarded by this said system enables the user to verify their documents once and then using that verification proof the users can open accounts or transact in other banks without any recurring verification. By utilising the blockchain as a shared trust layer rather than a primary data store, the system ensures that identity verification is separated from individual account on-boarding processes. This design preserves existing regulatory authority through central bank control while enabling seamless interoperability across the Bangladeshi banking sector.

The ecosystem is defined by four distinct logical roles that interact within the consortium. These roles are defined below:

Table 4.1: Logical Roles within the e-KYC platform

Role	Actor	Primary Responsibilities
Admin	Central Bank	Deploys smart contracts, manages network governance, and whitelists licensed financial institutions.
Issuer	Authorized Banks	Performs initial physical KYC verification and anchors cryptographic credentials to the ledger.
Verifier	Secondary Institutions	Consumes existing on-chain credentials to streamline customer onboarding without redundant checks.
Holder	Individual Customer	Maintains ownership of personal data and provides explicit cryptographic consent for third-party access.

The architectural data flow operates in three distinct phases to ensure efficiency and security. In the Issuance Phase an Issuer collects customer data off-chain, generates a cryptographic hash of the information (avoiding the storage of specific documents like NID, Driving License or photos on-chain) and anchors this hash along with essential metadata to the ledger. During the Verification Phase a Verifier discovers the on-chain hash and requests access; however, the system ensures that access to the underlying data is granted only after receiving a signed consent transaction from the User. In the final Revocation Phase, the original Issuer retains the authority to trigger an on-chain status update to "Revoked" if suspicious activity is detected or any crime is marked against their identity. An update is then instantly propagated to all Verifiers to maintain the AML compliance. Figure 1 illustrates this architectural data flow.

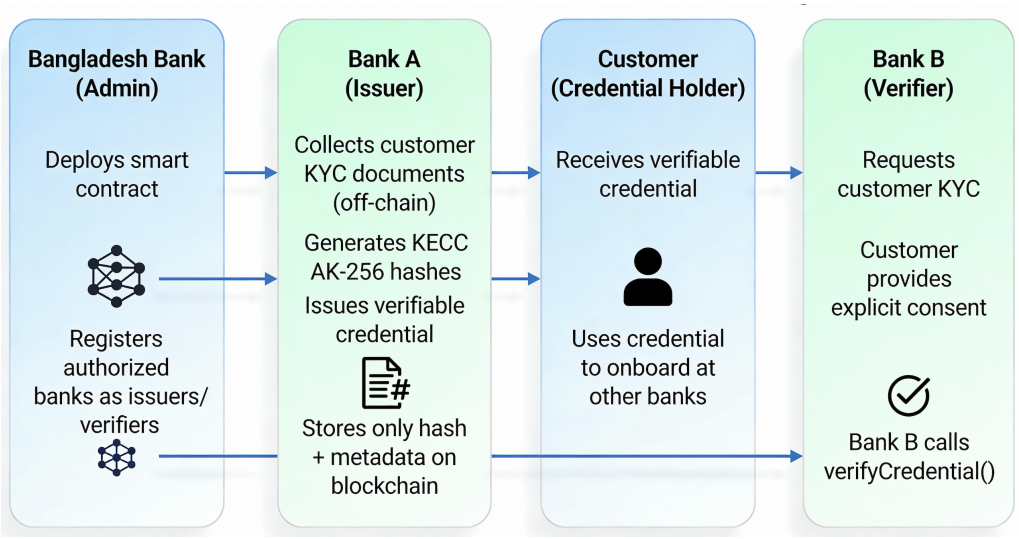


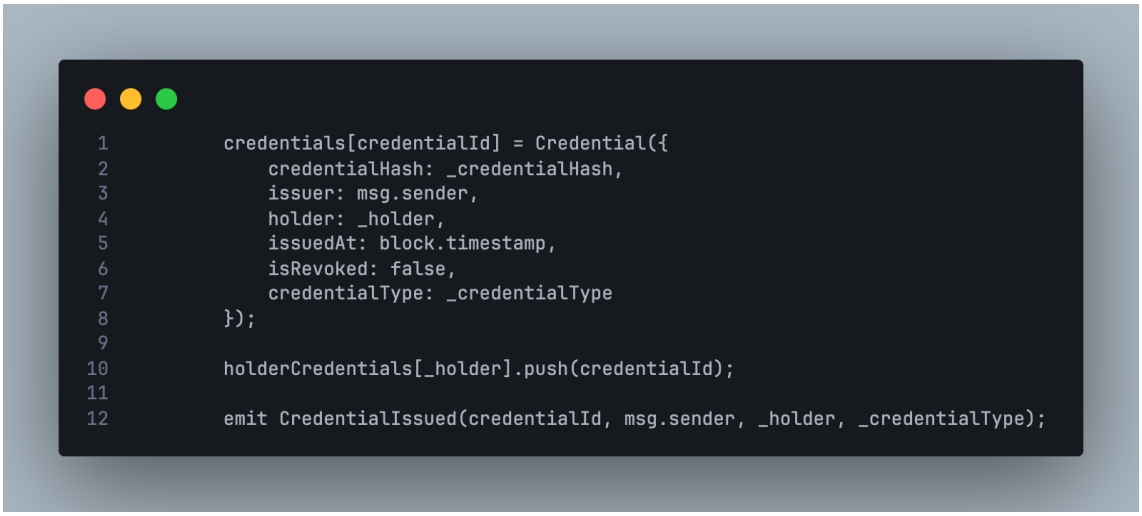
Figure 4.1: High-level system workflow illustrating the interaction between Admin, Bank A (Issuer), Customer (Credential Holder) and Bank B (Verifier) to facilitate one-time identity verification and cross-bank reuse.

```
const STATUS_TRANSITIONS = {
    draft: ['submitted', 'cancelled'],
    submitted: ['pending', 'under_review', 'approved', 'rejected', 'cancelled'],
    pending: ['under_review', 'approved', 'rejected', 'cancelled'],
    under_review: ['approved', 'rejected', 'additional_info_required', 'document_verification'],
    document_verification: ['under_review', 'additional_info_required', 'approved', 'rejected'],
    additional_info_required: ['under_review', 'approved', 'rejected', 'cancelled'],
    approved: ['issued', 'expired'],
    issued: ['expired'],
    rejected: [],
    expired: [],
    cancelled: []
};
```

Figure 4.2: The explicit state-transition matrix formalizes KYC lifecycle control and prevents invalid progression, making workflow behavior deterministic and auditable.

4.2 Gas Consumption and Performance Evaluation

The evaluation that was conducted by us on the sepolia testnet utilizing EIP-1559 dynamic fee mechanisms to improve fee predictability. We checked the industry standards that fluctuate depending on factors to lay out our performance and gas consumption evaluation. It is to be noted that these estimations change and are represented to show a fair estimation on the evaluation. To calculate our gains we used the formula gain = (industry average - our system / industry average) X 100 . The add authorization operation was evaluated to the operations that included authorization tasks related to blockchain ecosystems and we saw a 14-28% gain for this operation. And the issue credential saw a 8-26% improvement. These improvements occur because of the minimal SSTORE operations. SSTORE is an assembly level opcode on the Ethereum Virtual Machine (EVM) which is responsible for writing data to the blockchain's permanent storage. In this way by optimizing the state variables package and utilizing off-chain logs for non-critical data. With all this, the gas footprint is reduced overall.

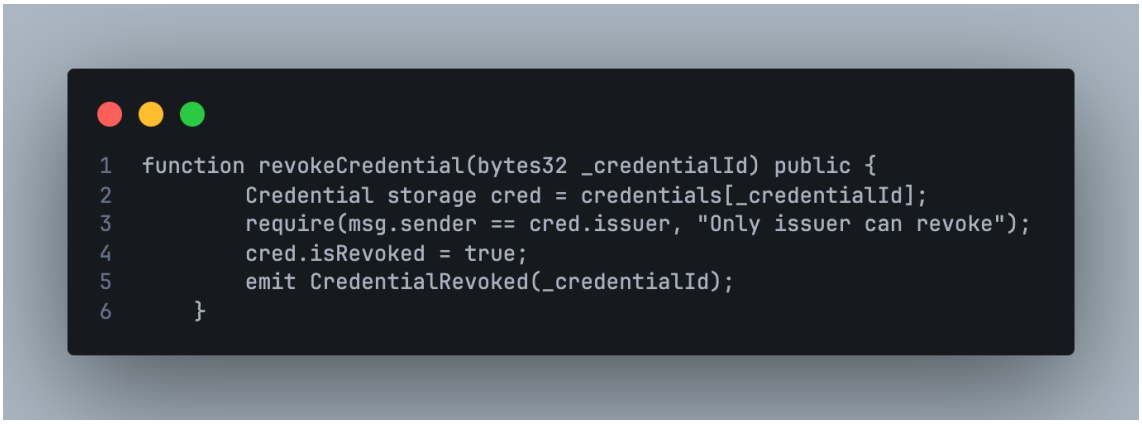


```
credentials[credentialId] = Credential({
    credentialHash: _credentialHash,
    issuer: msg.sender,
    holder: _holder,
    issuedAt: block.timestamp,
    isRevoked: false,
    credentialType: _credentialType
});

holderCredentials[_holder].push(credentialId);

emit CredentialIssued(credentialId, msg.sender, _holder, _credentialType);
```

Figure 4.3: This write operation stores only essential state and emits an event for traceability, supporting your argument about optimized on-chain storage cost.



```
function revokeCredential(bytes32 _credentialId) public {
        Credential storage cred = credentials[_credentialId];
        require(msg.sender == cred.issuer, "Only issuer can revoke");
        cred.isRevoked = true;
        emit CredentialRevoked(_credentialId);
    }
```

Figure 4.4: This minimal revocation logic performs a direct state flip and event emission, illustrating low-cost lifecycle invalidation for compliance updates.

Table 4.2: Performance Comparison with Industry Standards

Operation	**Our System (gas)**	**Industry Avg (gas)**	**Gain (%)**	**Rating**
Add Authorization	43,000	50,000–60,000	14–28%	Excellent
Issue Credential	184,000	200,000–250,000	8–26%	Very Good
Revoke Credential	32,000	35,000–45,000	9–29%	Excellent
Grant Consent	27,000	30,000–40,000	10–33%	Excellent
Verify (view)	0	0	Same	Standard

4.3 Results and Discussion

Like any new system, this platform comes with its own set of challenges that need to be honestly addressed.

To begin with, getting the green light from regulators like Bangladesh Bank and the Bangladesh Financial Intelligence Unit will not be simple. New rules or updated guidelines may be needed before the system can be officially used. On top of that, not every bank in Bangladesh is at the same level when it comes to technology. Many still rely on older systems, and connecting those to a modern blockchain platform could be tricky. Some banks might also be hesitant to make the switch, either because it would change how they normally do things or because they are not fully comfortable with the idea of sharing customer data across a shared platform.

Keeping that data safe is another concern. Storing sensitive information outside the blockchain in a secure and reliable way is not as straightforward as it sounds. There are also technical hurdles to think about, such as transaction fees on the Ethereum network, occasional slowdowns, and whether the system can hold up well as more and more users start using it over time.

Beyond the technical side, there are people and organizational challenges too. Bank staff who have spent years working with paper files and manual processes will need time and proper training to get comfortable with a fully digital system. This is not just a technical shift, it is a cultural one. Making the platform work in rural or less connected parts of the country adds another layer of difficulty, since people there may have limited access to smartphones or the internet, or may simply not be used to using digital services.

Finally, keeping everyone's information current across all participating banks will require strong coordination. When a customer's details change, or when access needs to be removed, every bank involved needs to be on the same page at the same time. That kind of smooth, ongoing cooperation will be one of the bigger tests for this system in practice.

Chapter 5

Impacts/Design Constraints of the Project

5.1 Impact of this Project on Societal, Safety, Legal and Economy

Bangladesh has a very large population. But many people still do not have a bank account. One big reason is how hard and slow the process of opening one is. Our platform makes that process much simpler. When verification becomes easy more people can join the banking system. This includes people in rural areas. It includes small traders. It includes people who have never had a bank account before. When more people get access to banking their quality of life improves. They can save money safely. They can send and receive payments easily. They can apply for loans when they need help. This is what financial inclusion looks like in practice. And our platform helps make it happen.

Identity fraud is a real and growing problem in Bangladesh. Fake documents and stolen identities cause serious damage. They hurt banks. They hurt innocent customers. And they are hard to catch with manual systems. Our platform stores a coded fingerprint of every verified document on the blockchain. Nobody can change that record quietly. Nobody can fake a verification that never happened. Bank officers can only access customer data after the customer gives clear permission. This protects people's private information. It also makes the whole banking process more trustworthy. Customers feel safer. Banks feel safer. And the system as a whole becomes more resistant to attacks and misuse.

Right now there is no single shared system that all banks in Bangladesh follow for identity verification. Each bank does things its own way. This creates gaps. It creates confusion. And it makes it harder for regulators to do their job. Our platform is built to follow the rules set by Bangladesh Bank. It also aligns with the guidelines of the Bangladesh Financial Intelligence Unit. Every action taken on the platform is recorded permanently. Regulators can check those records at any time. This makes it easier to enforce anti-money

laundering laws. It also makes it easier to track suspicious activity. Over time a system like this could push the entire banking sector toward better legal compliance. It could also serve as a foundation for future digital identity laws in Bangladesh.

Banks spend a lot of money on manual KYC processes every year. Staff time gets wasted on paperwork. The same checks get repeated at every bank for the same customer. Our platform cuts all of that repeated work. It brings down the cost of onboarding a new customer significantly. Banks save money. They can use those savings to offer better services. They can reach more customers. They can grow faster. Fraud also costs the economy a large amount of money each year. Because our records cannot be altered after they are saved fraud becomes much harder to commit. This protects both banks and customers. Faster onboarding also helps small business owners and traders who need quick access to financial tools. Overall this platform can lower banking costs. It can reduce fraud losses. And it can help Bangladesh move steadily toward a more digital and inclusive economy.

Chapter 6

Project Planning

This image is a Gantt chart showing the timeline for a project titled "Blockchain eKYC" running from August 2025 to April 2026 at North South University. It visually breaks down tasks, their durations, and overlaps across two courses: CSE499A and CSE499B.

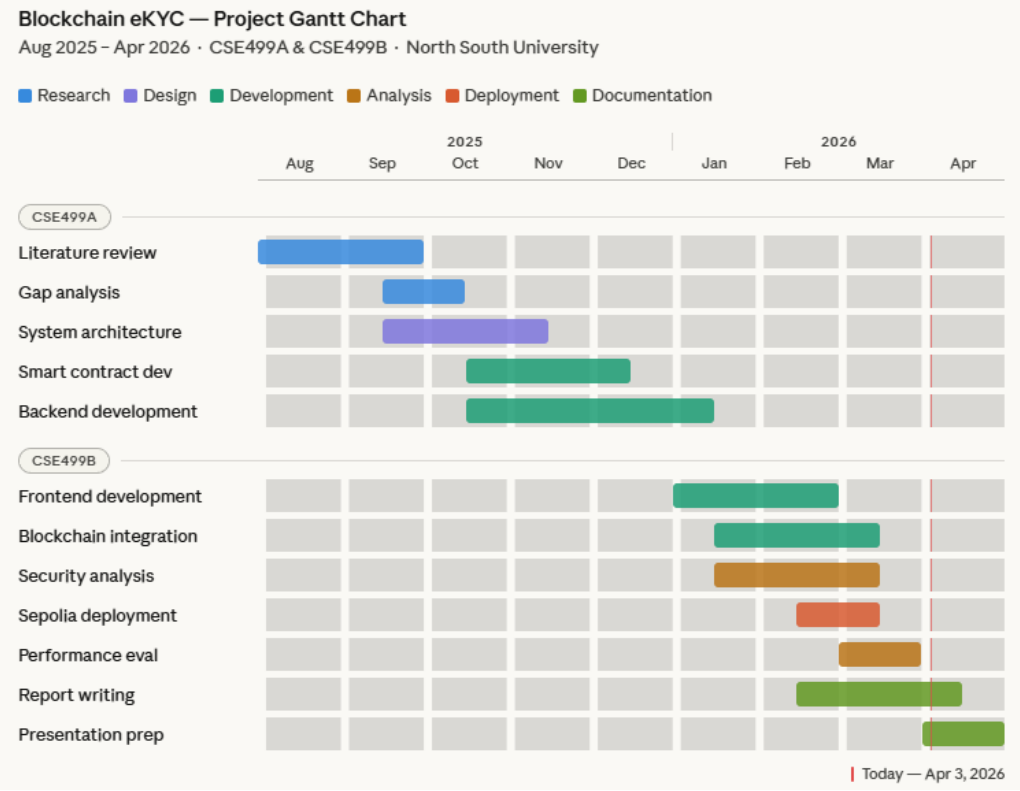


Figure 6.1: Gantt chart for project timeline

Chapter 7

Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

The problems that we were dealing with in the first hand are the matters related to how blockchain could be used to store the individual proof of verification in past scenarios. And even if that was the case how the separation of private data could be made with the proof of verification. Furthermore, there was the issue of how it could be established that the flow of proof and the matter of privacy preservation can be done without creating any substantial latency. The problem further extends to the fact that if the verification proof is public how can we design it in such a way that if there is consent from the user only then the credential proof can be reused.

Since there is a lot of preceding workflow present here it had to be made sure that the workflow is tamper proof. And as we are accommodating a lot of banks in the ecosystem it has to be made certain that the workflows are regulated considering the compliant regulations of the authorities.

Lastly it had to be made sure that the off chain integrity was preserved as well since personal documents had to be submitted there. And also it had to be made certain that upon documents verification the storage had to be cleaned so that these cannot be reused in any other way.

7.2 Complex Engineering Activities (CEA)

Our system is built by keeping a Privacy by Design framework in check. Data minimization principles were applied whenever sensitive information was involved. The system's operational mechanism relies on maintaining complete separation between personally identifiable information and the actual blockchain network verification proof(which is achieved after a successful verification of personal information for the beginning step). Any per-

sonal information remains encrypted within the private off-chain storage. And the actual verification proof is placed on the blockchain network. Through this the architecture stops data leakage while still permitting any authorized entity to confirm data integrity without fully disclosing any original records. On top of that upon a verification, the documents that were needed get removed from the system .

Consent management is embedded not as a superficial interface element but as a formal requirement within the system architecture. Upon consent from the customer or user access can be made possible for the bank officer's part to any verified credential. Each instance of granted consent action is recorded on the blockchain through a smart contract operation handled by backend logic. This obstructs unauthorized access or any form of silent data reuse. And at the same time the flow is fast considering just the system's scenario. If the consent is relayed as soon as possible there is no issue of any latency.

Furthermore, the system maintains a tamper-proof audit trail issuance, consent, and revocation activities. As these records cannot be retroactively modified this establishes a dependable compliance trail which is suitable for sound regulatory oversight.

Off-chain session integrity is preserved through a dedicated access control layer that operates in conjunction with the on-chain trust model. Backend services are secured using JSON Web Token based authentication. This enables stateless session management for both customers and bank officers. The JWT layer imposes strict API access controls to prevent unauthorized interactions with KYC endpoints. It bears emphasis that JWT serves as an off-chain security mechanism and does not supersede the cryptographic assurances provided by the blockchain ledger. The blockchain enforces integrity guarantees for the credentials themselves.

Chapter 8

Conclusion

In Bangladesh, most banks still check their customers' identities the old-fashioned way, doing everything by hand and repeating the same steps at every bank separately. This slows things down, wastes money, and opens the door to fraud. Some banks have tried to go digital, but without a proper system that everyone uses the same way, those efforts have not made much of a difference.

We are building a platform that uses blockchain technology to make this whole process better. It runs on Ethereum smart contracts, which help make sure that customer information stays safe, honest, and private. We did not store people's actual documents on the blockchain itself. Instead, only a coded fingerprint of those documents is saved there, while the real files are kept separately. This keeps personal information private while still creating a record that nobody can quietly change, and that regulators can check whenever they need to.

With this kind of system in place, banks can verify and bring in new customers much faster than before. Because records cannot be changed once they are saved, it also becomes much harder for anyone to commit fraud. Customers get a better experience, and banks no longer have to run the same checks over and over for the same person. The platform is also built to follow the rules set by Bangladesh Bank and the Bangladesh Financial Intelligence Unit, so it fits within the legal framework that already exists.

That said, there is still work to be done. More banks need to start using the system before it can reach its full potential. It should also be linked to national identity databases, and the way documents are stored off the blockchain needs to be made even more reliable. Still, what we have built shows that this kind of system can work in the real world. A blockchain-based e-KYC platform can offer a safer, clearer, and more efficient way to handle identity checks in Bangladesh. Over time, it could also help bring more people into the financial system, cut down on costs for banks, and build stronger trust between banks, customers, and regulators.

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